# Alex Wendland

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## Experience

**Software Engineer** Man Group 07/2023 - 01/2026

- Built an event-driven data quality framework on top of Kafka, Ray and Postgres with a front end in React. Adopted by 3 teams and deployed to over 50 datasets.
- Enabled annual database cost savings of £250K/year by migrating to a horizontally scaled Apache Arrow Flight server handling 1 GB/s.
- Drove initiative to triple datasets onboarding from 50 to 150 datasets per year by moving to a medallion data architecture backed by Iceberg and a low-code custom EL framework.
- Boosted fund chargeback by £80K/year by implementing Snowflake cost monitoring.
- Led a multi-team initiative to integrate LLM-powered code reviews in Bitbucket used by over 70 repos internally.

**Head of Data** CC Data 09/2021 - 06/2023

- Reduced Azure cloud storage costs by £60K/year (50%) through tiering policies, data format optimization, and retention strategies.
- Secured clients worth £60k/year by calculating historical and daily order book metrics on over 100TB of data using NumPy, Docker, and Azure Blob Storage.
- Expanded product range by researching, prototyping, and developing a blockchain data product to store, analyze, and standardize on-chain data.

**Science and Engineering Leadership Scheme** UK Government 10/2020 - 09/2021

- Managed high-profile scientific advice to the government in the Advisory Council for the Misuse of Drugs writing two 40+ page reports (e.g. Advice on consumer cannabidiol and Barriers to research).
- Facilitated social connection of remote civil servants using a JavaScript application backed by Google Sheets that scaled to four different organisations each with up to 600 users (still used as of 2024).

**Researcher and lecturer** University of Warwick 09/2015 - 09/2020

- Built a strong research portfolio, winning multiple academic awards, with publications in leading journals such as Advances in Mathematics, Annals of Applied Probability and the Journal of Graph Theory.
- Worked with Centre for Science and Policy (NLP), Heathrow airport (logistics modeling) and Defence Science and Technology Laboratory (AI safety).
- Award-winning teacher with experience in all levels and formats, from lecturing master's level courses to providing one-on-one support for underprivileged A-Level students.

## Education

**Masters in Computer Science** 4.0 GPA 09/2023 - Present
Georgia Institute of Technology, Atlanta, USA

**PhD in Mathematics** Generalisations of Cayley graphs and groups 09/2015 - 03/2020
University of Warwick, Coventry, UK

**Masters in Mathematics** 1st Class Honors (91% average) 09/2011 - 07/2015
University of Warwick, Coventry, UK

## Skills

**Programming Languages:** Python, SQL, Typescript, C, LaTeX, Nix.
**Tools and Frameworks:** Pandas, Numpy, SQLAlchemy, FastAPI, Pydantic, Apache Arrow, Kubernetes, Docker, Apache Iceberg, Snowflake, Azure Cloud, NLTK, ArcticDB, MongoDB, Grafana, Prometheus, OpenTelemetry, Sentry, Kafka, Postgres, Ray, React, NextJS.
**Collaboration and productivity:** JIRA, Bitbucket, Github, Confluence, Obsidian, Neovim, NixOS.